V93000 Specifications

V93000 Wave Scale RF-8 RFIM-8

Version 8.6.0

Published on 2023-02-24

Notices

Safety Notices

⚠ CAUTION A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

Warning A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

Wave Scale RF-8 with RFIM-8 specifications

The RF test solution with an RFIM-8 module attached to a WSRF-8 card provides the capabilities to generate and process RF signals with a frequency coverage from 10 MHz to 8 GHz.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

WS system calibration

WSRF-8 system calibration interval	Stim	Measure
3 months	0.2dB (typ) off vs calibration Note additional 0.15dB off in the first month for new produced cards	0.15dB (typ) off vs calibration Note additional 0.3dB off in the first month for new produced cards

Wave Scale RF-8 with RFIM-8 stimulus

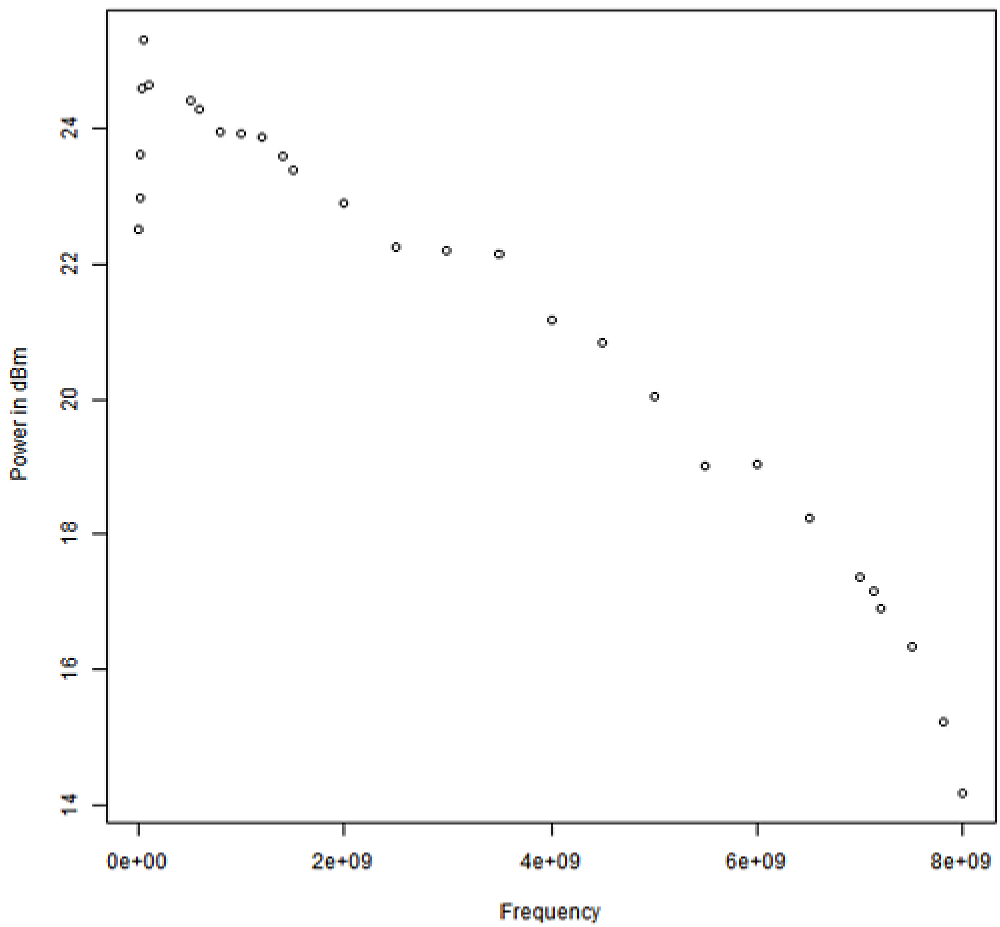
RFIM-8 stimulus maximum output power

Shows absolute maximum output power, measured as saturated power.

It is not recommended to use the Wave Scale Precision Source for power levels less than -80 dBm.

Carrier frequency (GHz)	Maximum output power (dBm) Spec	Maximum output power (dBm) Typ
0.011	18.4	22.8
0.016	19.6	23.3
0.031	20.6	24.3
0.051	21.2	24.9
0.1	20.7	24.3
1	20	23.8
2	18.9	22.7
2.5	18.4	22.2
3	17.9	21.8
4	16.7	20.5
5	16.4	20.3
6	15.1	19.3
7	13.3	16.8
7.125	12.3	16.7
7.5	10.7	15.8
8	9.7	14.2

Wave Scale RF-8 stimulus typical maximum power (average)



RFIM-8 stimulus modulation flatness

The measured performance for Wave Scale RF-8 stimulus modulation flatness is shown in the table below.

Carrier frequency range (GHz)	Maximum modulation bandwidth (MHz)	10 MHz	20 MHz	40 MHz	90 MHz	100 MHz	200 MHz
0.06 - 0.1	10	±0.15					
0.1 - 0.96	90	±0.15	±0.3	±0.5	±1.2		
0.96 - 3	200	±0.1	±0.2	±0.3		±0.8	±1.3
3 - 8	200	±0.1	±0.2	±0.4		±1.2	±1.7

RFIM-8 stimulus accuracy

Stimulus power accuracy

Power range (dBm)	-110…	-100…	-90…-	-80…-	-70…-	-60…-	-50…-	-40…-	-30…-	-20…-	-10…0	0…5

Frequency range (GHz)	0.05…	±0.73	±0.64	±0.56	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49
	1…2	±0.74	±0.66	±0.58	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52
	2…3	±0.76	±0.76	±0.67	±0.60	±0.53	±0.53	±0.53	±0.53	±0.53	±0.53	±0.53	±0.53
	3…4	±1.05 [1]	±1.14	±0.86	±0.62	±0.62	±0.62	±0.56	±0.56	±0.56	±0.56	±0.56	±0.56
	4…5	±1.21 [1]	±0.64 [1]	±0.87	±0.62	±0.62	±0.62	±0.56	±0.56	±0.56	±0.56	±0.56	±0.56
	5…6	±1.89 [1]	±0.79 [1]	±0.89	±0.65	±0.65	±0.65	±0.58	±0.58	±0.58	±0.58	±0.58	±0.58
	6…7	±1.77 [1]	±0.82 [1]	±1.11	±0.76	±0.76	±0.76	±0.62	±0.62	±0.62	±0.62	±0.62	±0.62
	7…7.5		±1.73 [1]	±1.21	±0.85	±0.78	±0.78	±0.65	±0.65	±0.65	±0.65	±0.65	±0.65
	7.5…8			±1.44 [1]	±1.12	±0.99	±0.91	±0.76	±0.76	±0.76	±0.76	±0.71	±0.76

RFIM-8 stimulus tone-combiner TOI (third-order intercept)

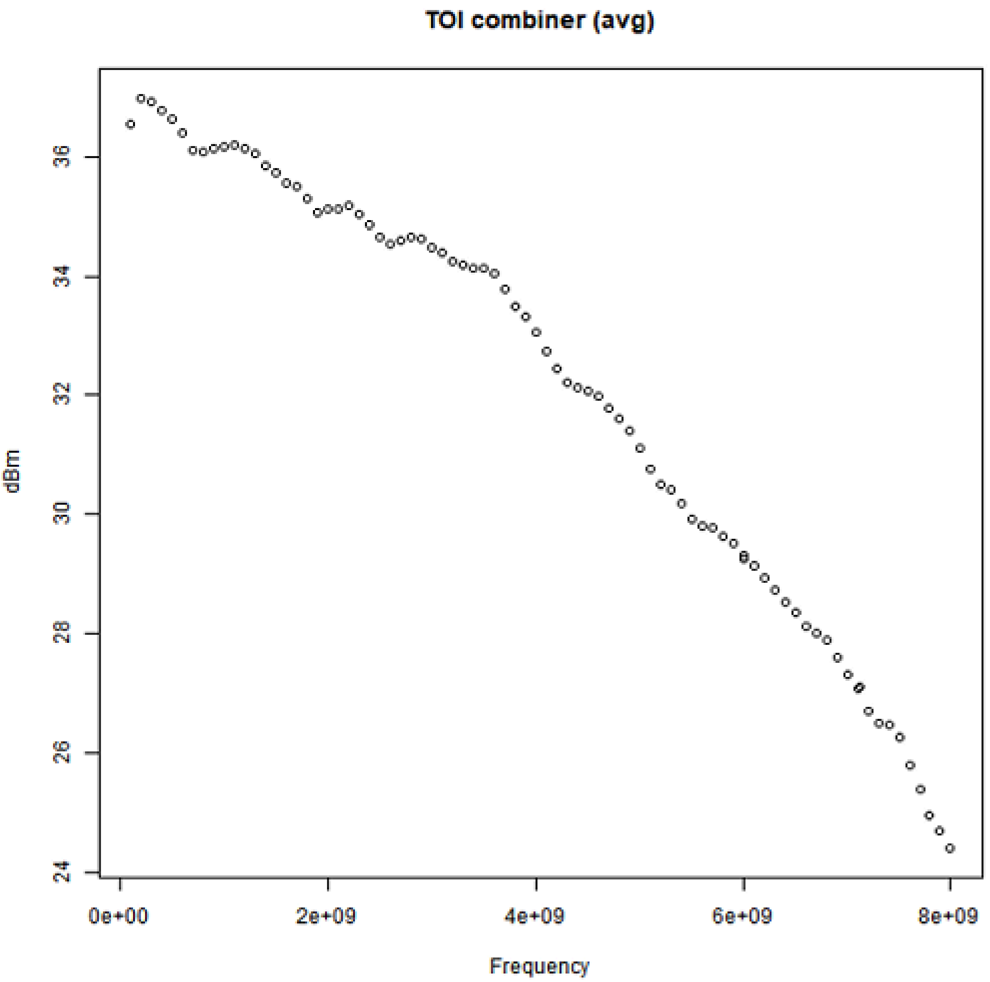
Conditions [2]: Tone-combiner mode, 100kHz tone spacing, optimizer balanced, 10 kHz Noise-Bandwidth.

Carrier frequency (GHz)	**Each tone power output at test port (dBm)**	**TOI (dBm) Spec**	**TOI (dBm) Typ**
0.1	0	30	36.2
0.2	0	30	36.8
0.5	0	30	36.4
1	0	29	36.0
2	0	29	34.9
3	0	28	34.2
4	0	27	32.6
5	0	25	30.8
6	0	23	29
7	0	21.6	27
7.5	0	19.6	25.7
8	0	17.6	23.9

[1] *Typical*

[2] *The maximum power delta between the two tones that the WSRF-8 card can guarantee is 60 dB for the above conditions.*

Wave Scale RF-8 stimulus typical TOI tone-combiner mode (average)



RFIM-8 stimulus modulated TOI (third-order intercept)

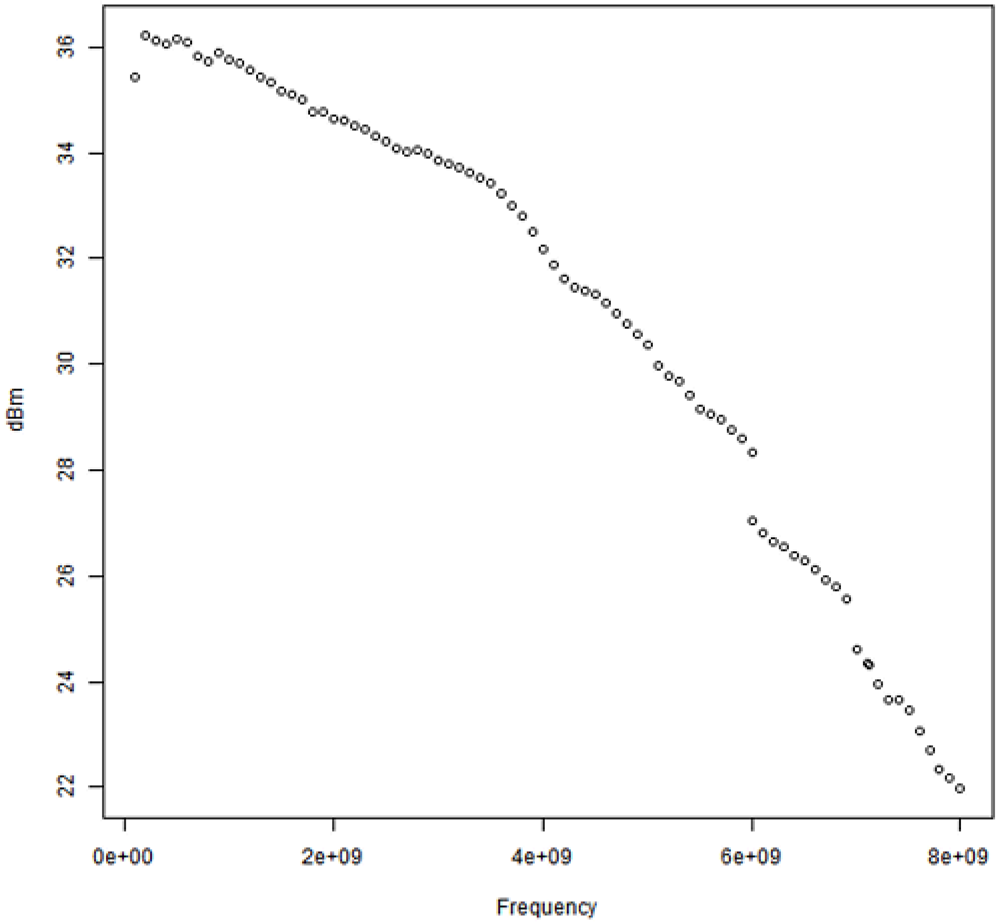
IQ-modulated 2-tone signal, tone-spacing 100kHz [3]

Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.5	0	29.3	35.4
1	0	29.3	35.1
2	0	27.8	34.2
3	0	27.8	33.5
4	0	25.8	31.7
5	0	23.8	30.0
6	0	21.6	26.6
7	-3	18.6	24.2
7.5	-3	16.6	23.0

[3] *2-tone IQ Signal, 250Msps, (tones at -10MHz, -10.1MHz, 0dBm or -3dBm each tone power), total power +3dBm or 0dBm, optimizer balanced, 10kHz Noise-Bandwidth*

Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
8	-3	14.6	21.5

Wave Scale RF-8 stimulus typical TOI modulated (average)



RFIM-8 stimulus harmonics

CW Pout≤0dBm, Optimizer mode IMD

Carrier frequency range (GHz)	Output power (dBm)	Harmonics, 2nd and above (dBc)	Subharmonics, 2nd (dBc)	Subharmonics, 3rd (dBc)
0.01-8	≤0	< -30	< -30	< -30

RFIM-8 stimulus phase noise

Specified CW stimulus phase noise (dBc/Hz), measured at 0 dBm stimulus power.

WSRF-8 stimulus phase noise [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-90	-104	-114	-116	-132
320	-90	-104	-114	-116	-132

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
545	-103	-117	-126	-130	-143
600	-103	-117	-126	-130	-143
885	-99	-113	-124	-127	-142
1900	-92	-107	-117	-119	-138
2400	-90	-106	-116	-119	-136
5800	-82	-98	-108	-111	-129
7000	-80	-96	-106	-109	-127

WSRF-8 + WSPS-8 stimulus phase noise [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-104	-119	-124	-132	-134
320	-104	-119	-124	-132	-134
545	-111	-120	-129	-142	-148
600	-111	-120	-129	-142	-148
885	-110	-120	-128	-142	-148
1900	-107	-119	-125	-141	-147
2400	-105	-118	-124	-141	-147
5800	-94	-115	-117	-138	-141
7000	-94	-112	-116	-135	-139

WSRF-8 + WSPS-8 stimulus phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-113.5	-124.9	-131.3	-139.7	-142.4
320	-113.5	-124.9	-131.1	-140.1	-143.3
545	-116.9	-126.2	-136.7	-146.9	-152.6
600	-116.9	-126.3	-136.8	-146.9	-152.5
885	-116.1	-125.5	-135	-146.3	-152
1900	-113.9	-124.9	-132	-145.7	-150.9
2400	-112.4	-124.5	-131.5	-145.5	-150.4
5800	-106.3	-119.4	-124.6	-141.7	-145.2

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
7000	-104.5	-117.7	-121.2	-139.5	-143.2

RFIM-8 stimulus random jitter

Measured at 0 dBm stimulus power.

WSRF-8 Stimulus random jitter

Frequency (MHz)	RMS jitter bandwidth	Random jitter (fs_{rms}) Typ	Random jitter (fs_{rms}) Max
600-7000	10 kHz to 10 MHz	107	173

Integration bandwidth : 10kHz to 10MHz

WSRF-8 + WSPS-8 stimulus random jitter max

Frequency [MHz]	Random jitter max [fs_{rms}]
600 <= f < 800	83
800 <= f < 1000	64
1000 <= f < 1400	53
1400 <= f < 1600	40
1600 <= f < 2000	36
2000 <= f < 2500	33
2500 <= f < 4000	31
4000 <= f < 5000	24
5000 <= f < 6000	21
6000 < f <= 7000	33

WSRF-8 + WSPS-8 stimulus random jitter typ

Frequency [MHz]	Random jitter typ [fs_{rms}]
600	54
885	41
1900	23
2400	19
5800	15
7000	18

RFIM-8 stimulus EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB, using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.8%	0.5%
5.6 (-15 dBm to -2 dBm)	0.8%	0.55%

Optimizer: balanced, NoiseBandwidth=20 MHz

802.11ac 160MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.9%	0.61%
5.6 (-15 dBm to -2 dBm)	0.9%	0.67%

Optimizer: balanced, NoiseBandwidth=10 MHz

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-25 to -6	0.77%	-42.27
	-5 < to 1	0.84%	-41.51

Optimizer: balanced, NoiseBandwidth=9 MHz

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-25 to -5	0.92%	-40.72
	-5 < to 1	0.95%	-40.45

Optimizer: balanced, NoiseBandwidth=20 MHz

WSRF-8 + WSPS-8 stimulus EVM

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM max	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.52%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB , using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-15 dBm to -2 dBm)	0.55%	0.4%
5.6 (-15 dBm to -2 dBm)	0.55%	0.42%

Optimizer: balanced, NoiseBandwidth=20 MHz

802.11ac 160MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-11 dBm to -2 dBm)	0.7%	0.53%
5.2 (-15 dBm to -11 dBm)	0.7%	0.54%
5.6 (-11 dBm to -2 dBm)	0.75%	0.56%
5.6 (-15 dBm to -11 dBm)	0.75%	0.57%

Optimizer: balanced, NoiseBandwidth=10 MHz

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-33 to < -30	0.81%	-41.83
	-30 to < -25	0.59%	-44.58
	-25 to < -16	0.46%	-46.74
	-16 to < -7	0.40%	-47.96
	-7 to < -2	0.45%	-46.94
	-2 to 0	0.54%	-45.35
	0 < to 3	0.72%	-42.85

Optimizer: balanced, NoiseBandwidth=9 MHz

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-30 to < -25	0.83%	-41.62
	-25 to < -18	0.66%	-43.61
	-18 to -5	0.61%	-44.29
	-5 < to < -2	0.65%	-43.74
	-2 to 0	0.68%	-43.35
	0 < to 3	0.86%	-41.31

Optimizer: balanced, NoiseBandwidth=20 MHz

RFIM-8 stimulus ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH, Center frequency: 2 GHz

Power level	ACLR [dBc]	ACLR2 [dBc]
-30 dBm to +4 dBm	-65.1 (typ)	-72 (typ)
-25 dBm to +1 dBm	-63 (max) [4]	-70 (max) [4]

Optimizer: balanced, NoiseBandwidth= 60 MHz, baseband attenuation 6 dB, baseband frequency offset 0 Hz

WSRF-8 + WSPS-8 stimulus ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, Center frequency: 2 GHz

Power level	ACLR [dBc]	ACLR2 [dBc]
-30 dBm to +4 dBm	-67.6 (typ)	-72.7 (typ)
-25 dBm to +1 dBm	-66.5 (max)	-71.5 (max)

Optimizer: balanced, NoiseBandwidth= 60 MHz, baseband attenuation 6 dB

RFIM-8 stimulus noise floor

CW 0 dBm, optimizer mode setting: noiseFloor, measured at +40 MHz offset relative to carrier frequency

Carrier frequency range (GHz)	Noise-floor (dBm/Hz)
0.05 - 0.545	-138
0.545 - 3.0	-142
3.0 - 5.0	-140
5.0 - 6.0	-138
6.0 - 8.0	-135

[4] Some baseband frequency offset values might result in dominant spur yielding higher ACLR than spec. To avoid this, change the basebandFrequencyOffset value.

RFIM-8 stimulus spurs

Measured at 0 dBm stimulus power

Note Frequencies defined in *RFIM-8 stimulus harmonics* on page 9 are excluded from this specification.

Valid for spur frequency offset ranging from +/-10kHz to +/-100MHz

WSRF-8 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-57
0.55	1.2	-65
1.2	1.7	-62
1.7	2	-60
2	2.05	-59
2.05	2.55	-60
2.55	3.1	-57
3.1	3.15	-53
3.15	3.4	-57
3.4	4	-54
4	4.1	-50
4.1	5.4	-53.5
5.4	5.45	-52
5.45	5.65	-54
5.65	6.9	-51
6.9	8	-50

Valid for spur frequency offset above +/-40MHz

WSRF-8 stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	2.5	-63
2.5	3.5	-60
3.5	4.5	-57
4.5	6	-50
6	7	-46

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
7	7.4	-44
7.4	7.8	-40
7.8	7.9	-38
7.9	8	-30

Valid for spur frequency offset ranging from +/-10kHz to +/-100MHz

WSRF-8 + WSPS-8 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-76
0.55	1.2	-80
1.2	1.8	-77
1.8	3	-73
3	6	-70
6	8	-68

Valid for spur frequency offset above +/-40MHz

WSRF-8 + WSPS-8 stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	6	-63
6	7	-63
7	7.4	-60
7.4	7.8	-54
7.8	8	-48

Wave Scale RF-8 with RFIM-8 measure

RFIM-8 measure power

Applicable for high resolution ADC only.

Applicable for manual path selection and attenuation settings.

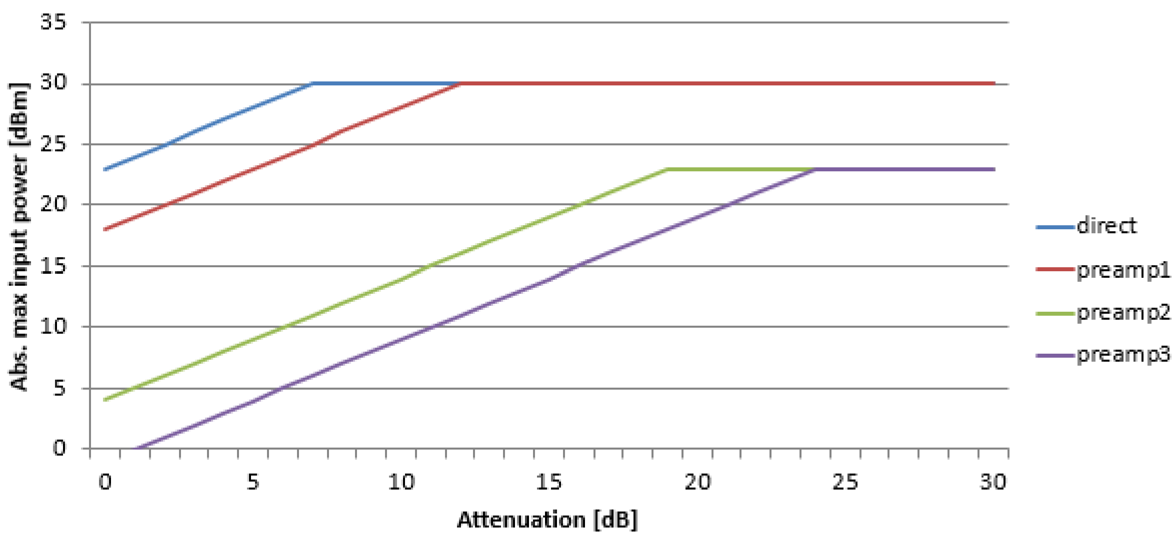
Note	The specified maximum RF input power values below are only valid for the splitch in switch mode.

Maximum RF-8 input power to all ports

Measure Path	**Receiver frequency range (CW) (GHz)**	**Maximum measurable power (dBm) with**		
		0 dB att spec	**0 dB att typ**	**30 dB att spec**
direct	0.01 to < 0.05	-4	-0.3	26
	0.05 to < 1	3	6.4	30
	1 to < 2	5	8.7	30
	2 to < 3	7	10	30
	3 to < 4	7	10.5	30
	4 to < 5	7	11.2	30
	5 to <6	9	12.1	30
	6 to 8	9	14.6	30
preamp1	0.01 to < 0.05	-13	-8.9	17
	0.05 to < 1	-6	-1.8	24
	1 to < 2	-4	0	26
	2 to < 3	-2	1.5	28
	3 to < 4	-1	2.2	29
	4 to < 5	-1	3	29
	5 to <6	0	3.4	30
	6 to 8	0	7.4	30
preamp2	0.01 to < 0.05	-21	-17.1	9
	0.05 to < 1	-16	-12.1	14
	1 to < 2	-14	-10.2	16
	2 to < 3	-12	-8.6	18
	3 to < 4	-12	-7.9	18

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
	4 to < 5	-11	-7	19
	5 to <6	-11	-6.3	19
	6 to 8	-11	-3.3	19
preamp3	0.01 to < 0.05	-30	-25.9	0
	0.05 to < 1	-24	-20.8	6
	0.2 to < 1	-22	-18.9	8
	2 to < 3	-21	-17.2	9
	3 to < 4	-20	-16.3	10
	4 to < 5	-19	-15.3	11
	5 to < 6	-18	-15	12
	6 to 8	-18	-10.9	12

Absolute maximum power damage level (dBm)



If the expected power value is larger than the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)", SmarTest will issue a warning.

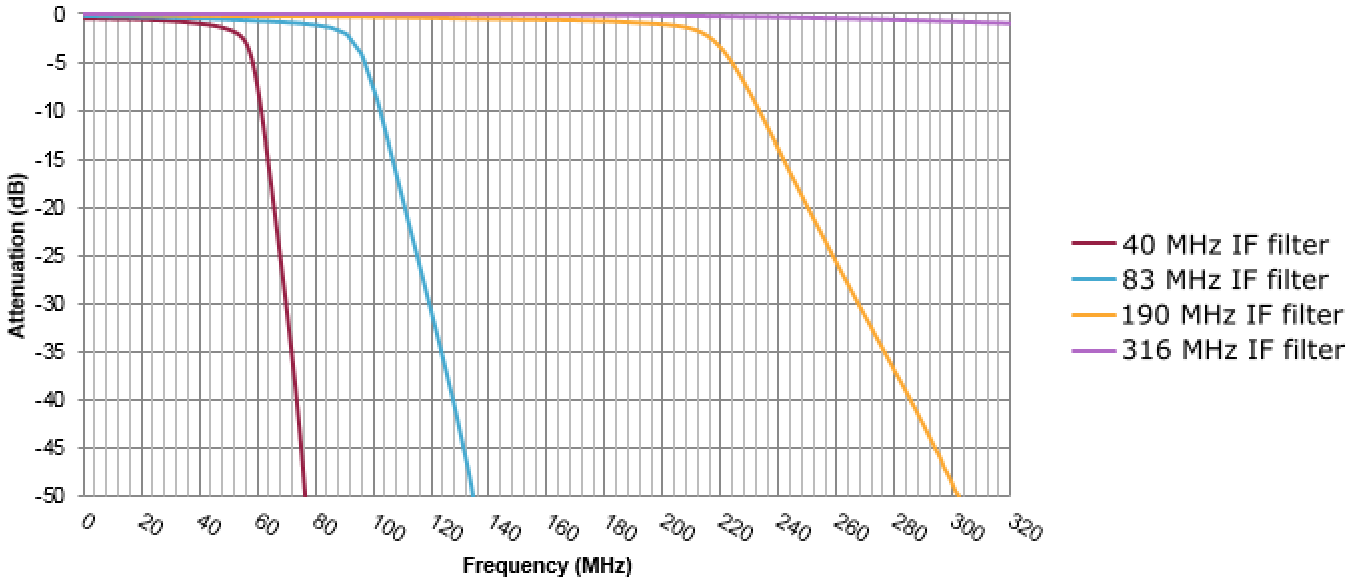
If the expected power value is larger than the "Absolute maximum power damage level(dBm)", SmarTest will issue an error.

Note When setting *rfmeas actions*, the following aspects should be taken into account:

- `expectedMaxPower` + `peakToAvgPowerRatio` < *Absolute maximum power damage level (dBm)* on page 18 (see figure above).
- `expectedMaxPower` + `peakToAvgPowerRatio` < *Maximum measurable power with 0dB att, spec(dBm)" + "Attenuation(dB)* (see table above)

- If the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)" is higher than "Absolute maximum power damage level(dBm)", the latter has priority.

Typical IF filter response within the RF path



RFIM-8 measure accuracy up to 8 GHz

Measure power accuracy conditions:

- Level accuracy is valid for CW signals in a 50 ohm system.
- Level accuracy is valid for High Resolution ADC, only.
- To achieve specified level accuracy for very low power levels appropriate bandwidth and result averaging must be applied.
- Specifications apply after a 30-minute warm-up and executions of all recommended WSRF calibrations.

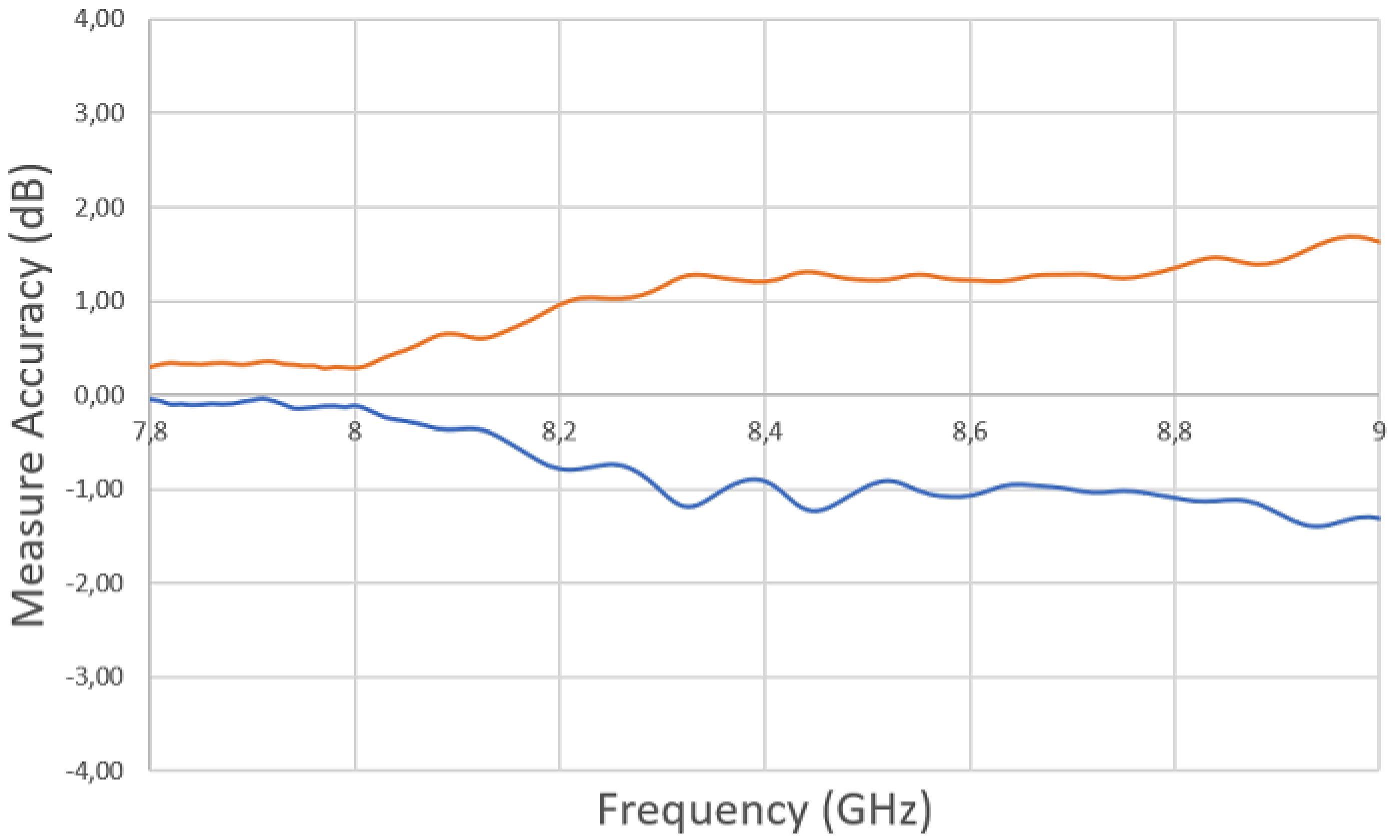
Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
0.050 ≤ f < 0.545	-1 ≤ Lv ≤ +20	±0.37
	-10 ≤ Lv < -1	±0.37
	-18 ≤ Lv < -10	±0.36
	-28 ≤ Lv < -18	±0.37
	Lv < -28	±0.37
0.545 ≤ f < 1.100	+3 ≤ Lv ≤ +20	±0.33
	-6 ≤ Lv < +3	±0.33
	-16 ≤ Lv < -6	±0.33
	-24 ≤ Lv < -16	±0.32
	Lv < -24	±0.32
1.100 ≤ f < 1.990	+4 ≤ Lv ≤ +20	±0.34

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
	-5 ≤ Lv < +4	±0.34
	-15 ≤ Lv < -5	±0.34
	-23 ≤ Lv < -15	±0.34
	Lv < -23	±0.34
1.990 ≤ f < 3.880	5 ≤ Lv ≤ +20	±0.40
	-4 ≤ Lv < 5	±0.40
	-14 ≤ Lv < -4	±0.40
	-22 ≤ Lv < -14	±0.40
	Lv < -22	±0.40
3.880 ≤ f < 5.400	+6 ≤ Lv ≤ +20	±0.44
	-3 ≤ Lv < +6	±0.43
	-13 ≤ Lv < -3	±0.43
	-21 ≤ Lv < -13	±0.43
	Lv < -21	±0.43
5.400 ≤ f ≤ 6.000	+8 ≤ Lv ≤ +20	±0.42
	-1 ≤ Lv < +8	±0.41
	-11 ≤ Lv < -1	±0.41
	-19 ≤ Lv < -11	±0.41
	Lv < -19	±0.41
6.000 < f ≤ 7.000	+10 ≤ Lv ≤ +20	±0.45
	+2 ≤ Lv < +10	±0.45
	-8 ≤ Lv < +2	±0.45
	-15 ≤ Lv < -8	±0.45
	Lv < -15	±0.45
7.000 < f ≤ 7.200	+10 ≤ Lv ≤ +20	±0.46
	+2 ≤ Lv < +10	±0.46
	-17 ≤ Lv < +2	±0.46
	-16 ≤ Lv < -7	±0.46
	Lv < -16	±0.46

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
7.200 < f ≤ 7.500	+10 ≤ Lv ≤ +20	±0.50
	+2 ≤ Lv < +10	±0.50
	-17 ≤ Lv < +2	±0.50
	-16 ≤ Lv < -7	±0.50
	Lv < -16	±0.50
7.500 < f ≤ 8.000	+10 ≤ Lv ≤ +20	±0.57
	+2 ≤ Lv < +10	±0.58
	-17 ≤ Lv < +2	±0.57
	-16 ≤ Lv < -7	±0.58
	Lv < -16	±0.58

RFIM-8 measure accuracy between 8 to 9 GHz

The measure accuracy for frequencies between 8 GHz and 9 GHz is shown below. This "measured" data represents 80% of the distribution of all measured curves.



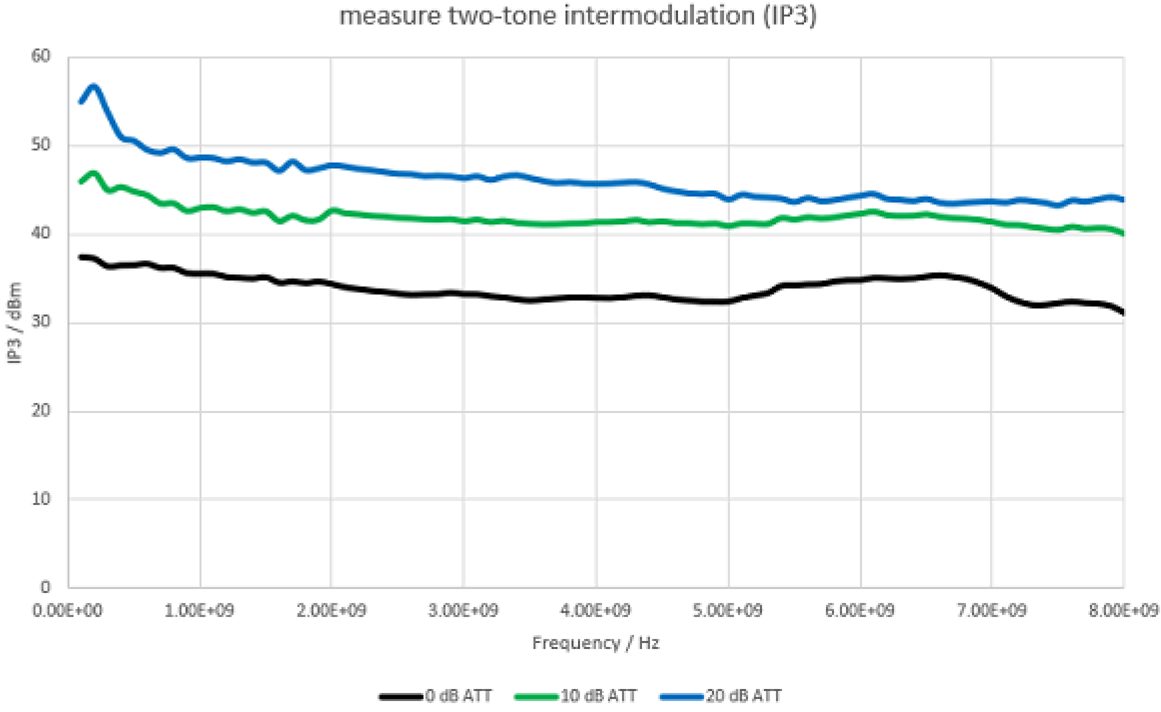
RFIM-8 measure two-tone intermodulation (IP3)

Conditions: HR ADC, 40 MHz filter, f_{IF} = 0.533 MHz, Δf = 100 kHz, -5 dBm tone power, with 0 dB receiver attenuation. IP3 increases with receiver attenuation.

Frequency: 10 MHz - 7.2 GHz

Measure Path	TOI (dBm)	
	Spec	**Typ**
direct	24	29.5
preamp1	15	20.9
preamp2	5	12.4
preamp3	-4	2.4

Measured data



RFIM-8 measure EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM spec	EVM typ
2 (-40 dBm to +2 dBm) Data obtained with loopback configuration	0.8%	
2 (-40 dBm to +10 dBm)		0.5%

802.11ac 80MHz

Frequency [GHz]	Power range [dBm]	EVM spec	EVM typ
5.2	-26 to -6	0.8%	0.54%
	-6 < to 0	0.8%	0.6%
5.6	-26 to -6	0.9%	0.58%
	-6 < to 0	0.9%	0.68%

802.11ac 160MHz

Frequency [GHz]	Power range [dBm]	EVM spec	EVM typ
5.2	-26 to -6	1%	0.67%
	-6 < to 0	1%	0.68%
5.6	-26 to -6	1.1%	0.7%
	-6 < to 0	1.1%	0.78%

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-5 to -6	0.77%	-42.27
	-5 < to 1	0.84%	-41.51

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-25 to -5	0.92%	-40.72
	-5 < to 1	0.95%	-40.45

WSRF-8 + WSPS-8 measure EVM

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM (max)	EVM (typ)
2 (-40 dBm to +10 dBm)	0.8%	0.47%

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM (max)	EVM (typ)
5.2 (-16 dBm to + 2 dBm)	1.0%	0.52%
5.2 (-26 dBm to -16 dBm)	1.0%	0.58%

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-33 to < -30	0.81%	-41.83
	-30 to < -25	0.59%	-44.58
	-25 to < -16	0.46%	-46.74
	-16 to < -7	0.40%	-47.96
	-7 to < -2	0.45%	-46.94
	-2 to 0	0.54%	-45.35
	0 < to 3	0.72%	-42.85

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-30 to < -25	0.83%	-41.62
	-25 to < -18	0.66%	-43.61
	-18 to -5	0.61%	-44.29
	-5 < to < -2	0.65%	-43.74
	-2 to 0	0.68%	-43.35
	0 < to 3	0.86%	-41.31

RFIM-8 measure ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC, Center frequency: 2 GHz

Power level	ACLR [dBc]
-16 dBm to +6 dBm	-63.0 (max)
-16 dBm to +6 dBm	-65.3 (typ)

Some IF value might result in dominant spur which yields to higher ACLR than spec. To avoid this, change IF value.

WSRF-8 + WSPS-8 measure ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC, Center frequency: 2 GHz

Power level	ACLR [dBc]
-16 dBm to +6 dBm	-67.0 (max)
-16 dBm to +6 dBm	-69.0 (typ)

RFIM-8 measure noise floor

Applicable for high resolution ADC only.

Measured at 0 dB receiver attenuation.

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
direct	0.01 to < 0.05	-130	-142
	0.05 to < 1	-138	-145
	1 to < 2	-136	-142
	2 to < 3	-135	-141
	3 to < 4	-135	-141
	4 to < 5	-133	-139
	5 to 6	-132	-137
	6 < to < 7	-132	-135
	7 to 7.2	-132	-136
	7.2 < to 8	-131	-135
preamp1	0.01 to < 0.05	-139	-150
	0.05 to < 1	-145	-152
	1 to < 2	-144	-150
	2 to < 3	-143	-148
	3 to < 4	-142	-148
	4 to < 5	-141	-146
	5 to 6	-139	-145
	6 < to < 7	-139	-143
	7 to 7.2	-140	-143
	7.2 < to 8	-138	-142
preamp2	0.01 to < 0.1	-140	-152
	0.1 to < 0.2	-152	-157
	0.2 to < 1	-152	-158
	1 to < 2	-152	-157
	2 to < 3	-152	-157
	3 to < 4	-151	-156

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
	4 to < 5	-151	-155
	5 to 6	-150	-154
	6 < to < 7	-149	-152
	7 to 7.2	-149	-152
	7.2 < to 8	-147	-151
preamp3	0.01 to < 0.1	-142	-152
	0.1 to < 0.2	-153	-158
	0.2 to < 1	-155	-159
	1 to < 2	-155	-159
	2 to < 3	-155	-159
	3 to < 4	-155	-159
	4 to < 5	-154	-158
	5 to 6	-154	-158
	6 < to < 7	-154	-156
	7 to 7.2	-153	-156
	7.2 < to 8	-152	-155

RFIM-8 measure dynamic range

Applicable for high resolution ADC only.

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
direct	0.01 to < 0.05	126	141.7
	0.05 to < 1	141	151.4
	1 to < 2	141	150.7
	2 to < 3	142	151
	3 to < 4	142	150.5
	4 to < 5	140	150.2
	5 to < 6	141	149.1
	6 to < 7	141	149.6
	7 to < 7.2	141	150.6

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
	7.2 to 8	140	149.6
preamp1	0.01 to < 0.05	126	141.1
	0.05 to < 1	139	150.2
	1 to < 2	140	150
	2 to < 3	141	149.5
	3 to < 4	141	150.2
	4 to < 5	140	149
	5 to < 6	139	148.4
	6 to < 7	139	150.4
	7 to < 7.2	140	150.4
	7.2 to 8	138	149.4
preamp2	0.01 to < 0.1	119	134.9
	0.1 to < 0.2	136	144.9
	0.2 to < 1	136	145.9
	1 to < 2	138	146.8
	2 to < 3	140	148.4
	3 to < 4	139	148.1
	4 to < 5	140	148
	5 to < 6	139	147.7
	6 to < 7	138	148.7
	7 to < 7.2	138	148.7
	7.2 to 8	136	147.7
preamp3	0.01 to < 0.1	112	126.1
	0.1 to < 0.2	129	137.2
	0.2 to < 1	131	138.2
	1 to < 2	133	140.1
	2 to < 3	134	141.8
	3 to < 4	135	142.7
	4 to < 5	135	142.7

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
	5 to < 6	136	143
	6 to < 7	136	145.1
	7 to < 7.2	135	145.1
	7.2 to 8	134	144.1

RFIM-8 measure spurs

Measured at 0 dBm stimulus power

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF-8 measure spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max Spur [dBc]
0.13	1	-67
1	1.2	-64
1.2	1.6	-61.5
1.6	1.7	-54
1.7	3.4	-56.5
3.4	4	-53
4	4.1	-51
4.1	5.15	-53
5.15	5.25	-52.5
5.25	5.9	-50.5
5.9	6.9	-51
6.9	8	-50

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF + WSPS measure spurs (measured values)

Input frequency from [GHz]	Input frequency to [GHz]	Max spurs [dBc]
0.13	4	-70
4	6	-68
6	8	-67

Wave Scale RF-8 with RFIM-8 terminate

RF ports that are turned-off, have an internal 50-Ohm termination to GND. In order to avoid thermal damage to the termination resistors, the following abs-max restrictions have to be observed per logical board (half of RFIM).

Number of ports in termination mode	Maximum power applied per port (dBm)
1 or 2	30
3 to 9	27
10 or more	26